Bounding VIO Drift with Self-Supervised Aerial Geo-Localization

GNSS-denied UAV navigation, CPU-only

Nijat Abbasov

August 28, 2026

Two estimators, opposite failure modes

VIO — smooth, drifts without bound
700 m over 3.6 km = 19% of distance flown

Geo-localization — absolute, never drifts
but only 36% of predictions are correct

Fuse them.
VIO gives shape.
Geo-localization gives scale and position.

MUN-FRL `bell412_dataset4`
3.6 km, Ottawa International Airport
CPU only — no GPU on the airframe

	OpenVINS	Fused
Real-time RMSE	489 m	**248 m**
Best run	464 m	**198 m**

−49%
mean of three runs

−57%
best run

CAEVL — geo-localization without paired data

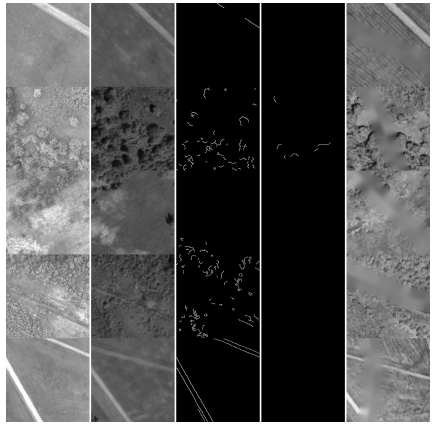
Domain gap: photo now vs. satellite imagery months old.

Fix: discard appearance — both sides become **Canny edge maps**.

Trains on reference imagery only:

1. Autoencoder → 1024-d descriptor
   + DINOv2 perceptual loss
2. VICReg on (clean, degraded) pairs
   no negatives, no paired UAV data

⇒ new terrain needs **satellite tiles only**



ViLD reproduction	R@1
Paper	49.68%
Ours	48.14%
+ perceptual loss	**52.88%**

OpenVINS — and the dataset problem

One camera + an IMU. It never recognises places it has already been, so errors only accumulate.

From the takeoff point it starts **14.7 m** out and ends **700 m** out after 3.6 km.

The *shape* is right, but the path comes out **too small** — as if the flight were shrunk. Stretching it back fixes most of the error.

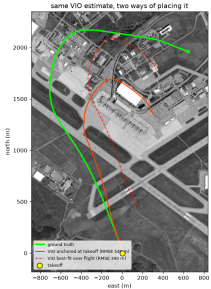
Why so few usable datasets?

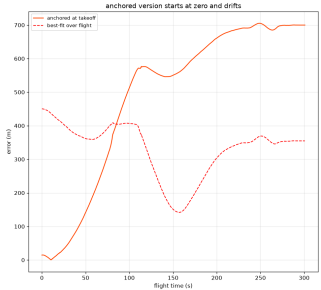
Camera tracking wants to fly **low**

things move more between frames

Map matching wants to fly **high**

each photo covers more ground





Dataset	Altitude	IMU
VPAIR	300 m	no
AerialVL	—	no
SatLoc	100–300 m	not released
Zurich MAV	5–15 m	too low
MUN-FRL	**25–170 m**	**yes**

Which predictions can we trust?

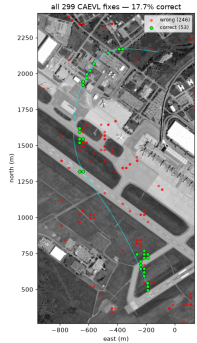
Only **36%** of matches are right — and the network's own confidence is *higher when it is wrong*:

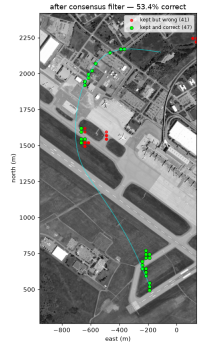
confidence	correct	wrong
flight 1	0.9429	**0.9493**
flight 2	0.9447	**0.9598**

So we ignore confidence entirely.

Instead: keep only matches that **agree with each other**, once you account for how far the aircraft moved between them.

Wrong matches land in random places — two agreeing is a 0.5%





	before	after
matches	299	101
% correct	17.7%	**99.0%**
good ones kept	107	100 **(93%)**

Causal fusion

$$\hat{\mathbf{p}}_k = s\,\mathbf{R}(\theta)\,\mathbf{p}_k^{\text{vio}} + \mathbf{t} \quad \text{— estimated online, } \textbf{from past predictions only}$$

On each new prediction

1. enters a buffer
2. *trailing* window screened for consensus
3. survivors accumulate
4. refit $(s, \theta, \mathbf{t})$ to all accepted history

Before handover: dead-reckon on VIO from the known takeoff pose
$\Rightarrow$ position output at *every* instant

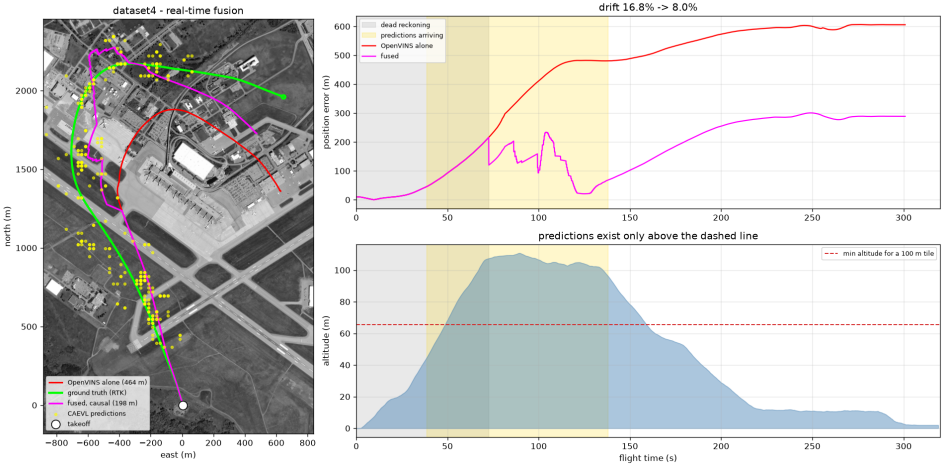
Why estimate s and θ?

The system starts not knowing which way is north, and the VIO path comes out too small — both must be learned from the matches.

An EKF lost: 422 m vs 347 m

Scale error grows *linearly*, process noise as $\sqrt{\cdot}$. The filter turns overconfident and gates out correct predictions — most certain when most wrong.

Result — real-time, full flight



Red: OpenVINS. Green: RTK truth. Magenta: fused. Grey: dead reckoning before handover

at $t{+}73\,\mathrm{s}$. **464 m** $\rightarrow$ **198 m**

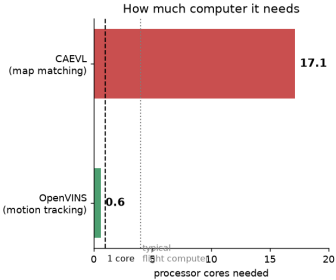
Summary — accuracy

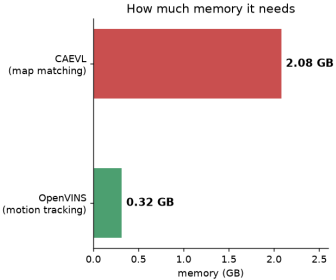
	OpenVINS alone	+ **Geo-loc**	Improvement
Position error (RMSE)	489 ± 44 m	$\mathbf{248 \pm 86}$ **m**	-49%
best run	463.8 m	**198.2 m**	-57%
Drift over the flight	$17.7 \pm 1.5\%$	$\mathbf{10.1 \pm 3.6}\%$	-43%
Heading error	*cannot be measured — this dataset has no reference heading*		

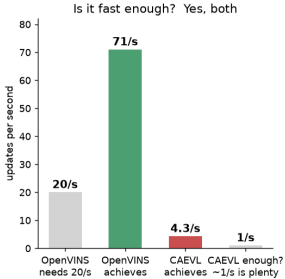
Drift is the final position error as a fraction of distance flown: **after 3.6 km the aircraft ends up 10% of that distance away from where it really is**, instead of 18%.

Mean of three OpenVINS runs on identical inputs. Both trajectories start from the known takeoff point; the system uses only information available at the time.

Summary — can it run on the aircraft?







Motion tracking is cheap
under one core, 0.3 GB, and 3.5× faster than the camera needs

Map matching is the expensive part
17 cores here. On a single core it would give one fix every 4 s — still useful, since fixes are only needed occasionally.

Limitations and next steps

Availability, not accuracy, is the limit

- 100 m tile needs ∼66 m altitude
  only **34%** of this flight qualifies
- Below that: 1–5% correct
  adding them made fusion *worse*: 347 → 527 m
- Smaller tiles collapse the descriptors
  2.21% of pairs > 0.99 similar, vs 0.31% at 100 m
- Altitude and initial heading still from ground truth
  deployment: altimeter + magnetometer

Next

1. **Query-side scale normalisation**
   one database at the lowest operating altitude; rescale queries down to it
2. **Retrain on map tiles the right size**
   trained on tiles covering 1438 m of ground, used on 100 m — a 14× mismatch
3. Find out why the VIO path comes out too small
   also the source of run-to-run spread: 540 vs 464 m from identical inputs

`github.com/NijatAbbasov06/uav-geolocalization`